Final Report

**Title: Smart Gardening System Using Esp32 and Blynk IOT**

# 1. Introduction

Smart gardening is an innovative approach that uses modern technology to automate and optimize plant care. With the increasing global focus on sustainable practices, smart irrigation systems have become essential for conserving water, improving crop yield, and reducing manual effort. Traditional gardening requires constant human supervision to monitor soil conditions, water levels, and nutrient balance. This often leads to challenges such as over-watering, under-watering, and improper soil health management. These issues directly impact plant growth and overall productivity. To address these problems, the integration of IoT (Internet of Things) in gardening has emerged as a highly effective solution.

The Smart Gardening System using ESP32 is designed to automate the irrigation process and monitor soil conditions in real time. The ESP32 microcontroller, equipped with built-in Wi-Fi, enables seamless data communication with the Blynk IoT platform, allowing users to monitor sensor readings and control pumps remotely from their smartphones. The system incorporates two key sensors: a soil moisture sensor, which detects the water content of the soil, and a pH sensor, which measures the acidity or alkalinity of the soil. These sensors provide crucial information that helps maintain optimal growing conditions for plants.

The system controls two water pumps using a 2-channel relay module. Pump 1 automatically activates when soil moisture drops below a threshold (e.g., 50%), ensuring plants receive adequate water. Pump 2 is controlled based on soil pH conditions, allowing corrective actions when the soil becomes too alkaline. In addition, the Blynk interface provides a manual mode, enabling users to operate the pumps independently whenever needed.

This IoT-based smart gardening solution reduces labour, minimizes water wastage, and promotes healthy plant growth. It is cost-effective, easy to build, and suitable for home gardens, greenhouses, nurseries, and agricultural fields. With its combination of automation, monitoring, and remote control, the system demonstrates how IoT can transform traditional gardening into a smarter and more efficient process.

# 2. Objectives of the Project

The primary objective of the Smart Gardening System using ESP32 is to create an intelligent, automated, and efficient solution for maintaining healthy plant growth with minimal human intervention. As traditional gardening requires continuous monitoring of soil moisture, pH levels, and irrigation schedules, it often results in excessive water usage, manual labor, and inconsistent plant care. This project aims to address these challenges by integrating sensors, automation, and IoT technology into a unified smart system.

One major objective is to automatically monitor soil moisture using a soil moisture sensor. By continuously analyzing the moisture content of the soil, the system ensures that plants receive water only when necessary. This prevents conditions such as over-watering and under-watering, which are common causes of root damage and poor plant health. Through automation, the system maintains an ideal moisture level without requiring manual checks.

Another key objective is to measure and regulate soil pH levels using a pH sensor. Soil pH plays a significant role in nutrient availability and plant growth. By monitoring pH values, the system can activate a second pump to correct alkalinity conditions when required. This feature provides a more holistic approach to soil management by addressing both water content and chemical balance.

A further objective is to enable remote monitoring and control through IoT technology. By integrating the ESP32 microcontroller with the Blynk IoT platform, users can observe real-time data such as moisture levels, pH values, and pump status directly on their smartphones. This remote-access feature enhances convenience and allows users to manage their garden from any location.

Additionally, the system aims to provide dual operational modes: Automatic and Manual. In Automatic Mode, pumps operate based on predefined threshold values. In Manual Mode, the user can control the pumps independently using Blynk switches. This flexibility allows for human oversight when needed while still benefiting from automation.

Another important objective is to reduce water wastage and promote sustainable gardening practices. By activating pumps only when required, the system significantly minimizes unnecessary water consumption. This makes the solution eco-friendly, cost-effective, and ideal for regions facing water scarcity.

Finally, the project aims to develop a scalable and low-cost system that can be expanded with additional sensors such as temperature, humidity, or NPK sensors in the future. This makes it suitable for home gardening, greenhouses, agricultural fields, and smart farms.

# 3.Components Used

## Hardware Components

**1. ESP32 Development Board**

A powerful microcontroller with built-in Wi-Fi used to read sensors, process data, and control pumps.

Acts as the central controller for automation and IoT communication via Blynk.

**2. Soil Moisture Sensor**

Measures the water content in the soil by detecting resistance or capacitance changes.

Helps decide when Pump 1 should turn ON or OFF automatically.

**3. pH Sensor Module**

Detects the acidity or alkalinity of the soil using voltage variations.

Provides data to maintain healthy soil conditions by activating Pump 2 when needed.

**4. 2-Channel Relay Module**

Works as an electronically controlled switch to operate the two pumps safely.

Isolates the high-voltage pump circuitry from the low-voltage ESP32 pins.

**5. DC Water Pumps**

Pump 1 irrigates plants based on moisture level; Pump 2 adjusts soil pH conditions.

Controlled using relay signals for automatic or manual operation.

**6. External Power Supply (5V/12V)**

Powers the pumps and relay module, as ESP32 cannot drive them directly.

Ensures stable operation without overloading the microcontroller.

**7. Breadboard & Jumper Wires**

Used for making temporary circuit connections during prototyping.

Helps connect sensors, relays, and ESP32 in a structured way.

## Software Requirements

**1. Arduino IDE**

A programming platform used to write and upload code to the ESP32 board.

Helps integrate libraries, sensors, and IoT functions easily.

**2. Blynk IoT Application**

Mobile app that displays sensor values and allows remote control of pumps.

Connects to the ESP32 via the cloud for real-time monitoring.

**3. ESP32 Board Manager**

Adds ESP32 board support to Arduino IDE for compiling and uploading code.

Enables access to ESP32-specific features like Wi-Fi and ADC functions.

**4. Wi-Fi Network**

Provides internet connectivity for ESP32 to communicate with Blynk cloud.

Enables remote monitoring and control from anywhere in the world.

# 4. System Overview

The Smart Gardening System is an *IoT-based automated plant care solution* designed to monitor soil conditions and control irrigation without human effort. The system uses the ESP32 microcontroller as the main controller, which continuously reads data from two sensors: a soil moisture sensor and a pH sensor. These sensors help determine the water content and acidity level of the soil.

Based on the sensor readings, the ESP32 decides whether to turn ON or OFF two different water pumps connected through a 2-channel relay module.

- Pump 1 waters the plants when the soil is too dry (moisture below 50%).
- Pump 2 operates when the soil pH becomes too high (pH above 8), helping maintain optimal soil balance.

The entire system is connected to the Blynk IoT platform, which allows users to monitor moisture, pH values, and pump status in real time using a mobile app. Users can also manually control both pumps from anywhere through the internet.

This system ensures efficient water usage, improves plant health, reduces manual work, and makes gardening smarter and more convenient.

# 5. Working Principle

**Step 1: Sensor Reading**

- Moisture sensor → detects soil water content
- pH sensor → measures alkalinity or acidity

**Step 2: Processing**

ESP32 reads sensor values through ADC pins and converts them into meaningful data.

**Step 3: Decision Making**

ESP32 checks the moisture and pH values against threshold limits.

**Step 4: Pump Control**

- If soil is dry → Pump 1 ON
- If soil pH is high → Pump 2 ON

Control signals are sent to the **relay module**, which switches the pumps.

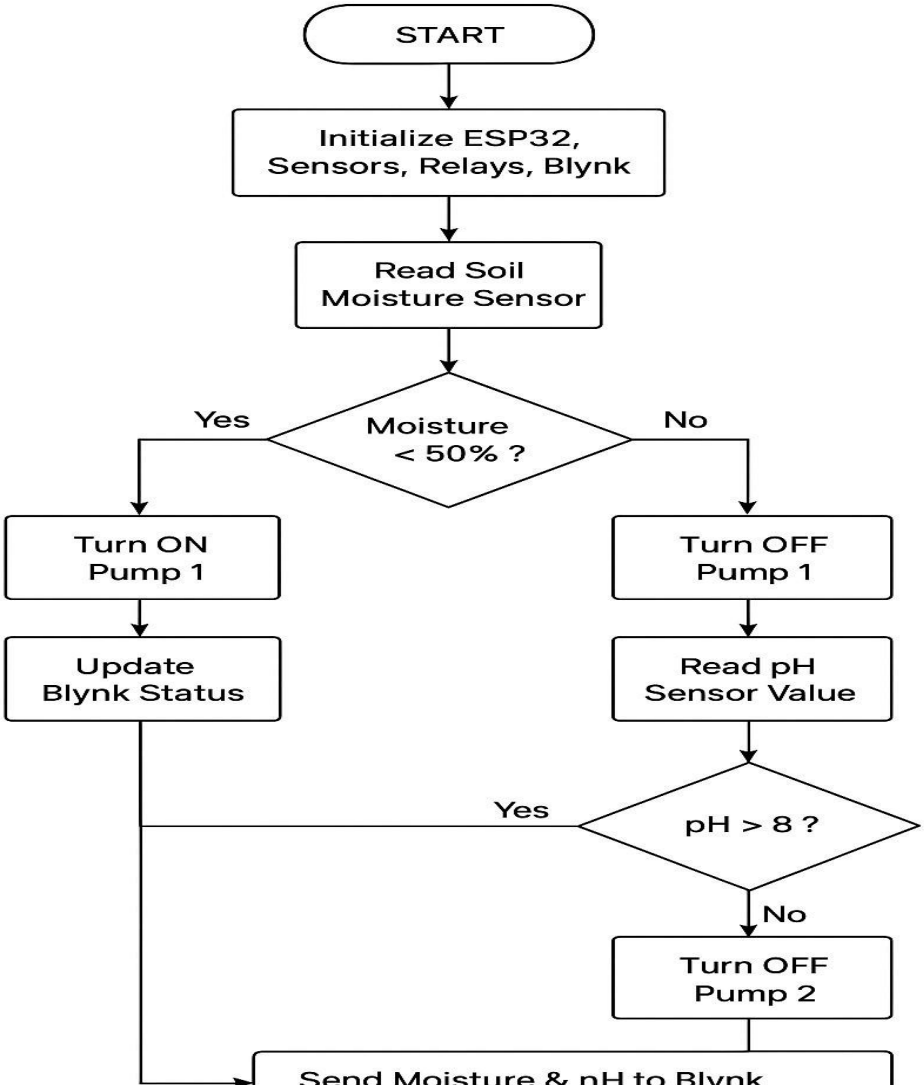
**Step 5: IoT Monitoring**

All data is uploaded to the Blynk cloud, allowing remote monitoring. **Step**

**6: Manual Control Option**

The user can override automatic mode using Blynk mobile interface.

# 6.Flowchart of Working




**Flowchart Explanation**

1. Start the system
2. Initialize ESP32, sensors, relays, and Blynk
3. Read moisture sensor
4. If moisture < 50%, turn ON Pump 1
5. Else turn OFF Pump 1
6. Read pH sensor
7. If pH ≥ 8, turn ON Pump 2
8. Else turn OFF Pump 2

9. Send data to Blynk
10. Repeat

# 8.Circuit Description

The circuit of the Smart Gardening System is built around the ESP32 microcontroller, which serves as the central processing unit for reading sensor data and controlling the pumps through a relay module. Each component is connected in a way that ensures accurate sensing, safe switching, and stable power distribution.

**1. Sensor Connections**

Two analog sensors are used in the system:

- Soil Moisture Sensor
- pH Sensor Module

Both sensors provide analog output, so their signal pins are connected to the ADC (Analogto-Digital Converter) pins of the ESP32. For example:

- Moisture Sensor → GPIO 34
- pH Sensor → GPIO 35

The VCC of the moisture sensor is connected to 3.3V, while the pH sensor module often requires 5V, and both share a common GND with the ESP32. This common ground is important for stable analog readings.

**2. Relay Module Connections**

A 2-channel relay module is used to operate two DC water pumps. Relay pins are connected as follows:

- Relay IN1 → GPIO 26 (controls Pump 1)
- Relay IN2 → GPIO 27 (controls Pump 2)
- Relay VCC → External 5V
- Relay GND → Common ground with ESP32

The relay acts as an electronic switch, allowing the ESP32 to turn pumps ON/OFF without directly supplying high current to them. This protects the ESP32 from power overload.

**3. Pump Connections**

Each pump is wired through one relay channel.

- Pump positive terminal → external 5V/12V supply

- Pump negative terminal → Relay NO (Normally Open) terminal
- Relay COM (Common) terminal → Power supply ground

When the relay is activated by ESP32, the NO and COM terminals close the circuit and the pump turns ON.

**4. Power Supply Management**

The ESP32 runs on USB or 5V, but pumps require higher current, so an external power supply (5V or 12V depending on pump) is used. To avoid malfunction or noise:

- All grounds (ESP32, relay, sensors, power supply) are connected together.
- Pumps never draw power directly from ESP32.

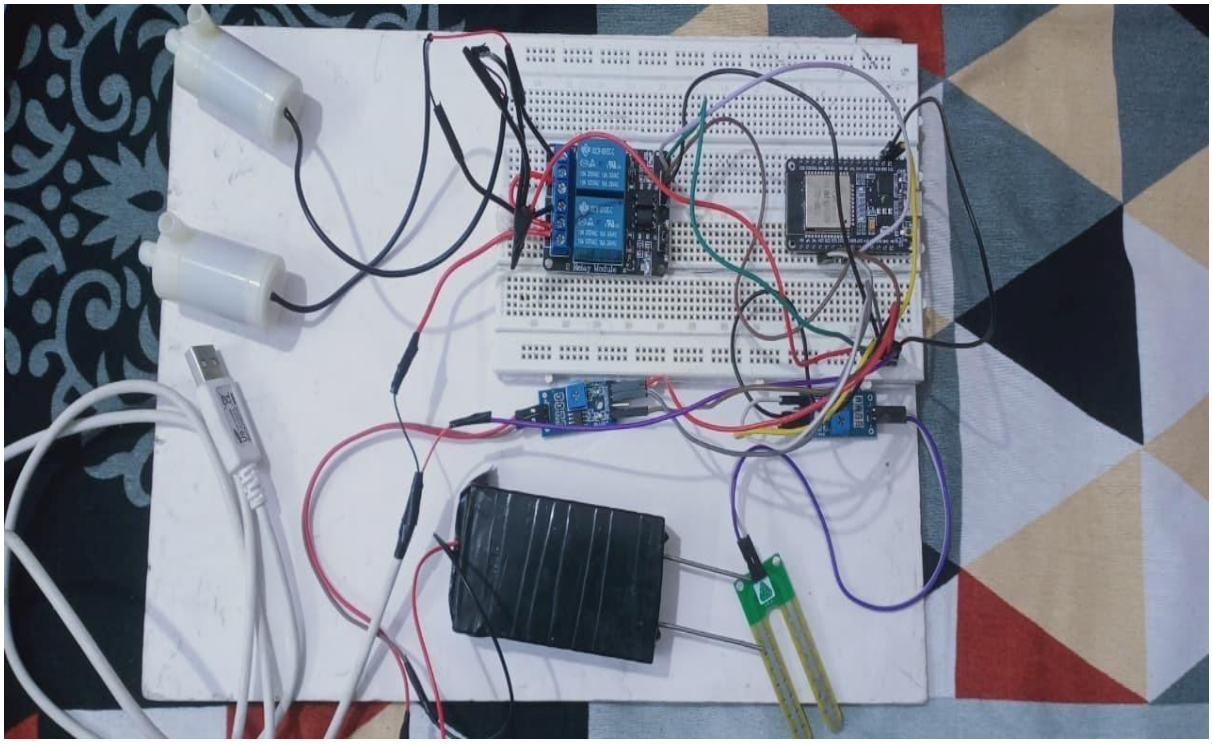
**5. IoT Communication**

The ESP32 is programmed to connect to Wi-Fi, and all sensor readings and relay status values are uploaded to the Blynk IoT cloud. This enables remote monitoring and control using a smartphone.

**6. Overall Circuit Function**

- Sensors send analog data → ESP32
- ESP32 processes values and decides pump ON/OFF
- Relay receives digital control signals
- Pumps operate safely using external power
- Data is continuously displayed on Blynk

This integrated circuit ensures automation, safety, and real-time monitoring for efficient gardening.

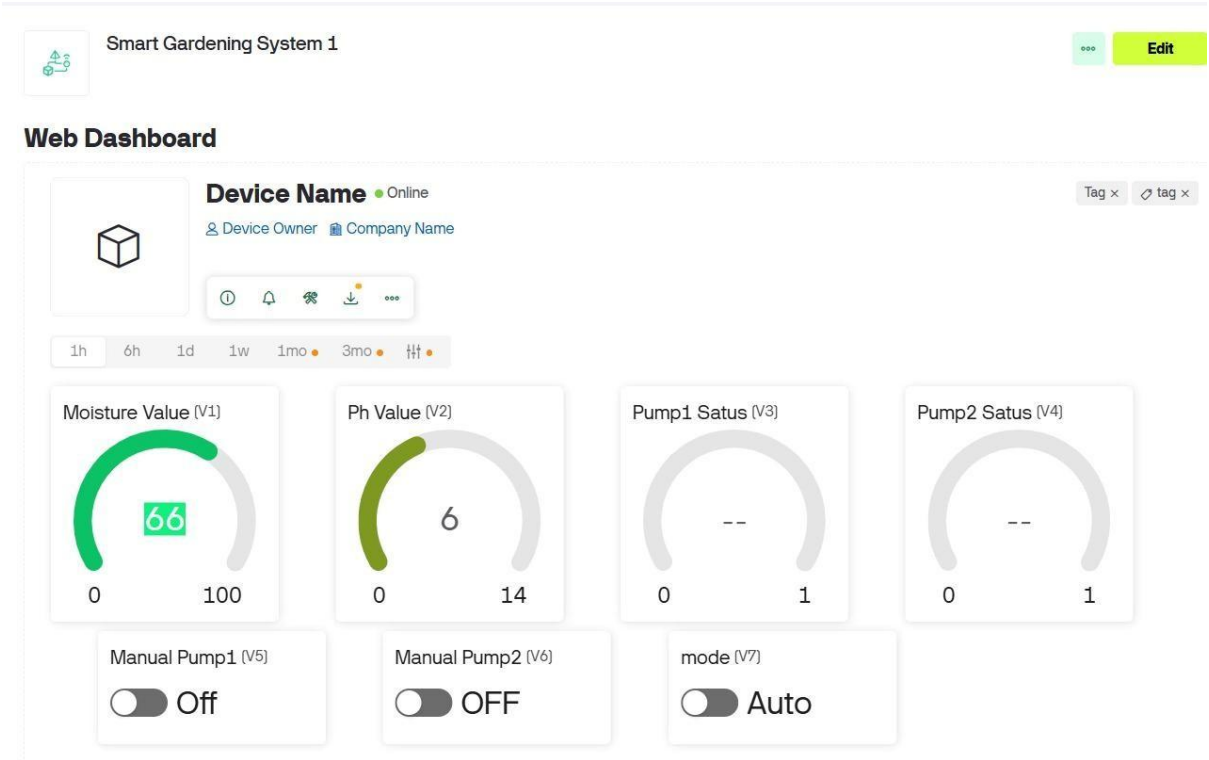


# 9.Blynk Interface Description

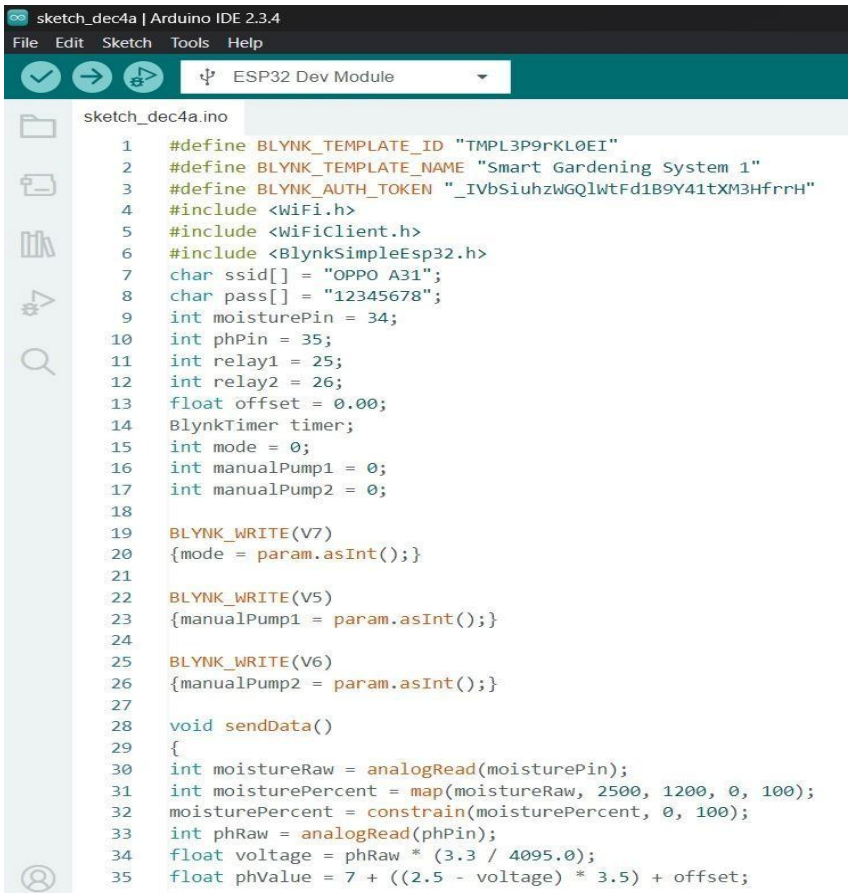
The Blynk App dashboard includes:

- **Moisture Gauge (V1)**
- **pH Gauge (V2)**
- **Pump 1 Status (V3)**
- **Pump 2 Status (V4)**
- **Manual Pump 1 Control (V5)**
- **Manual Pump 2 Control (V6)**
- **Auto/Manual Mode Switch (V7)**

This allows the user to monitor and control the entire system remotely.




# 10.Code




```
#define BLYNK_TEMPLATE_ID "TMPL3P9rKL0EI"
#define BLYNK_TEMPLATE_NAME "Smart Gardening System 1"
#define BLYNK_AUTH_TOKEN "_IVbSiuhzWGQlWtFd1B9Y41tXM3HfrrH"
#include <WiFi.h>
#include <WiFiClient.h>
#include <BlynkSimpleEsp32.h>
char ssid[] = "OPPO A31";
char pass[] = "12345678";
int moisturePin = 34;
int phPin = 35;
int relay1 = 25;
int relay2 = 26;
float offset = 0.00;
BlynkTimer timer;
int mode = 0;
int manualPump1 = 0;
int manualPump2 = 0;

BLYNK_WRITE(V7)
{mode = param.asInt();}

BLYNK_WRITE(V5)
{manualPump1 = param.asInt();}

BLYNK_WRITE(V6)
{manualPump2 = param.asInt();}

void sendData()
{
int moistureRaw = analogRead(moisturePin);
int moisturePercent = map(moistureRaw, 2500, 1200, 0, 100);
moisturePercent = constrain(moisturePercent, 0, 100);
int phRaw = analogRead(phPin);
float voltage = phRaw * (3.3 / 4095.0);
float phValue = 7 + ((2.5 - voltage) * 3.5) + offset;
```

sketch_dec4a | Arduino IDE 2.3.4

File Edit Sketch Tools Help

ESP32 Dev Module

sketch_dec4a.ino

```
if (mode == 0)
{
if (moisturePercent < 50) {digitalWrite(relay1, LOW); Blynk.virtualWrite(V3, 1);}
else {digitalWrite(relay1, HIGH); Blynk.virtualWrite(V3, 0);}
if (phValue < 8.0) {digitalWrite(relay2, HIGH); Blynk.virtualWrite(V4, 0);}
else {digitalWrite(relay2, LOW); Blynk.virtualWrite(V4, 1);}
}
else if (mode == 1)
{
if (manualPump1 == 1) {digitalWrite(relay1, LOW); Blynk.virtualWrite(V3, 1);}
else {digitalWrite(relay1, HIGH); Blynk.virtualWrite(V3, 0);}
if (manualPump2 == 1) {digitalWrite(relay2, LOW); Blynk.virtualWrite(V4, 1);}
else {digitalWrite(relay2, HIGH); Blynk.virtualWrite(V4, 0);}
}

Blynk.virtualWrite(V1, moisturePercent);
Blynk.virtualWrite(V2, phValue);
Serial.print("Moisture: "); Serial.println(moisturePercent);
Serial.print("pH: "); Serial.println(phValue);
}

void setup() {
Serial.begin(115200);
pinMode(relay1, OUTPUT);
pinMode(relay2, OUTPUT);
digitalWrite(relay1, HIGH);
digitalWrite(relay2, HIGH);
Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
timer.setInterval(2000, sendData);
}

void loop() {
Blynk.run();
timer.run();
}
```

# 11.Results

The Smart Gardening System was tested under real environmental conditions, and the overall performance showed highly successful outcomes. The system accurately measured both soil moisture and pH values using the connected sensors. The readings were stable, consistent, and responded quickly to changes in soil conditions, ensuring reliable monitoring for healthy plant growth.

The automated pump control worked exactly as intended. Pump 1 activated whenever the soil moisture dropped below the set threshold (50%), and stopped as soon as the moisture level increased, preventing over-watering. Pump 2 responded correctly to pH readings, turning ON only when the soil became too alkaline. This automation reduced manual effort and ensured plants always received what they needed at the right time.

The Blynk IoT integration functioned smoothly throughout testing. Manual control buttons for both pumps responded instantly, allowing users to override the automatic mode whenever necessary. Sensor values refreshed in real time on the mobile app, ensuring continuous visibility of the garden’s condition.

With the intelligent watering mechanism, the system significantly improved water usage efficiency. Since pumps ran only when required, unnecessary water flow was eliminated.

This not only conserved water but also protected plants from issues like root damage caused by excess watering.

Overall, the results demonstrate that the system is stable, efficient, and highly effective for real-world gardening applications. It provides a reliable automated solution while still giving users full control through IoT technology.

# 12.Advantages

**1. Automated watering system**

The system automatically activates the water pump whenever the soil moisture falls below a preset level. This ensures plants receive the right amount of water at the right time without human intervention, reducing the chances of over-watering or under-watering.

**2. Saves water and reduces human effort**

By supplying water only when required, the system prevents wastage and optimizes water usage. It also eliminates the need for regular manual checking and watering, making gardening effortless and efficient.

**3. Real-time IoT monitoring**

The integration with the Blynk IoT app allows users to monitor soil moisture, pH levels, and pump status from anywhere in the world. This provides full visibility and control over the system using a smartphone.

**4. Low-cost and easy to implement**

The project uses easily available components like ESP32, sensors, relay modules, and pumps, making it cost-effective. The circuitry and programming are simple, making the system easy to build even for beginners.

**5. Ensures plant health and growth**

By maintaining proper moisture and pH levels, the system creates an ideal environment for plants to grow. Healthy soil conditions improve nutrient absorption and overall plant development.

# 13. Applications

**1. Home gardening**

Ideal for maintaining indoor plants, balcony gardens, and small home gardens without frequent manual care.

**2. Agriculture & greenhouses**

Automates irrigation and maintains soil health in large-scale farming or controlled greenhouse environments, improving productivity.

**3. Nurseries**

Helps nursery owners maintain thousands of plants consistently by automating watering based on real-time soil needs.

**4. Smart farming**

Fits perfectly into modern precision agriculture techniques where IoT and automation are used to monitor and optimize crop growth.

**5. Soil analysis systems**

Moisture and pH monitoring together help in understanding soil conditions, making it useful in research, soil testing labs, and agricultural education.

# 14. Conclusion

The Smart Gardening System using ESP32 successfully demonstrates how IoT can transform traditional gardening into an efficient and automated process. By integrating sensors such as the soil moisture sensor and pH sensor with a relay-based pump control mechanism, the system ensures plants receive optimal water and soil conditions at all times. The use of the Blynk IoT platform brings real-time monitoring and manual control to the user's smartphone, making the system highly user-friendly.

This project highlights the potential of IoT in agriculture by reducing water waste, improving plant growth, and minimizing human dependency. With its scalable design, low cost, and easy implementation, the system can be applied to home gardens, farms, nurseries, and greenhouses. Overall, it provides a practical and modern solution to improve gardening efficiency and sustainability.

# 15. Future Enhancements

**1. Adding temperature & humidity sensing**

Integrating an environment sensor (DHT11/DHT22) would allow monitoring of air temperature and humidity, giving more control over the plant's atmosphere and helping adjust irrigation patterns.

**2. Automated fertilizer mixing using NPK sensors**

Adding NPK (Nitrogen, Phosphorus, Potassium) sensors would enable the system to analyze soil nutrients. A fertilizer pump could be automated to maintain balanced soil nutrition.

**3. Solar-powered system**

Using a solar panel and battery backup can make the system energy-efficient and suitable for outdoor farming in remote areas with no electricity.

**4. AI-based water prediction**

Machine learning can be used to analyze past data, weather conditions, and plant types to predict future watering needs, making the system intelligent and self-learning.

**5. Web dashboard monitoring**

Along with the mobile app, a full web-based dashboard can be developed for remote monitoring through laptops and PCs, useful for farmers and greenhouse operators.